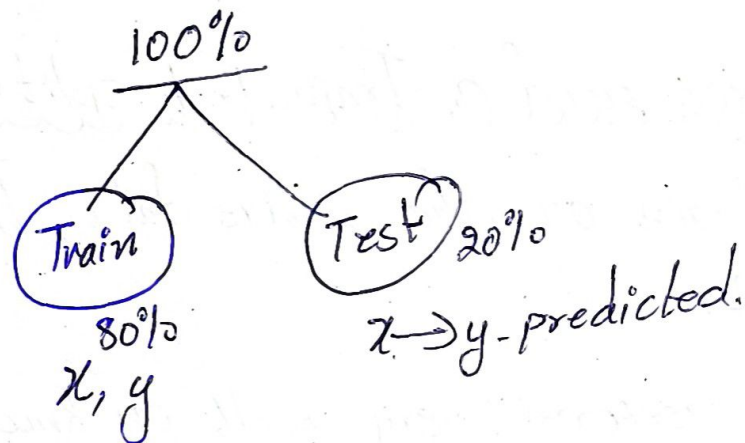


Model Evaluation Validation & Data Validation:-

[train-test-split]

→ From model-selection we need to import the train-test-split.



When we build a machine learning model, we must check how well it performs on new, unseen data.

To do this, we split the dataset into two parts:

1. Training Data (Train Set)

- Usually 70-80% of the data
- Used to teach the model patterns
- The model learns the relationship b/w input (X) and output (y).

2. Testing Data (Test Set):

- Usually 20-30% of the data.

→ used to Evaluate the model

→ This data is not shown to the model during training.

→ So, we can check how well it predicts new data.

Why do we need a Train-Test split?

→ If we train on the entire data, the model memorizes it.

→ It will perform very well on known data but poorly on new data.

→ This problem is called overfitting.

Train-test split helps us:-

- Test the model on unknown data
- know how the model behaves in real world.

How Train-Test Split works:-

Suppose you have 100 rows of data.
you split it as:-

80 rows → training

20 rows → testing

Workflow:-

1. Train the model on 80 rows
2. Predict on the remaining 20 rows
3. Compare actual vs predicted values
4. Calculate metrics like RMSE, accuracy, R^2 , etc.

Suppose 90% of trained & 60% of testing then it is overfitting.

If 60% of trained & 50% of testing then it is underfitting.

Example:-

```
import sklearn as sdr
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x, y,
                                                    test_size=0.2, random_state=16)
x_train, x_test, y_train, y_test # Execute here.
```

After training & testing model:-

```
from sklearn.linear_model import LinearRegression
sdr = LinearRegression()
sdr.fit(x_train, y_train) → Execute
```

$y_train_pred = sdr.predict(x_train)$

from sklearn.metrics import r2_score

$r2_score(y_train, y_train_pred)$

O/P :- 0.9682256656957373

$y_test_pred = sdr.predict(x_test)$

$r2_score(y_test, y_test_pred)$

O/P :- 0.7949890259295